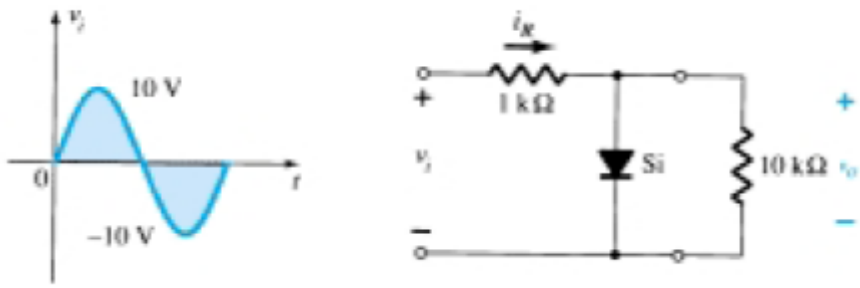
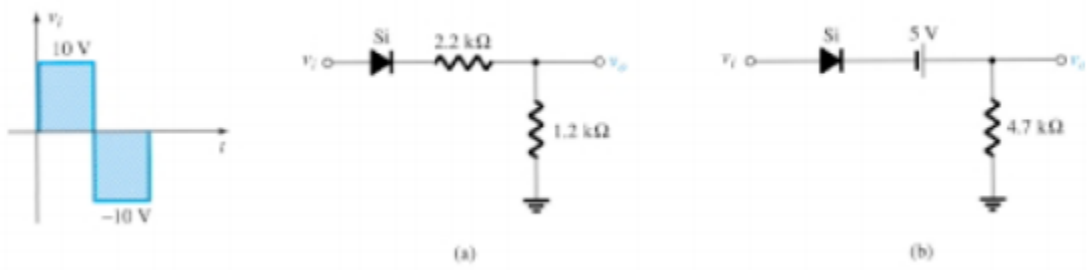
UNIT 1-QUESTION BANK

1. Using diode eqn. determine the diode current at 20°C for a silicon diode with Is = 50 nA and an applied forward bias of 0.6 V.

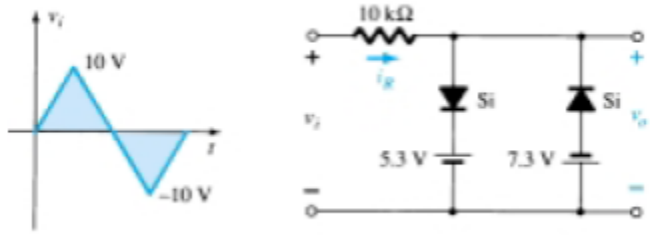
2.For the network of Fig. below, sketch vo and iR



3. Determine vo for each network of Fig. below for the input shown.



4. Sketch iR and vo for the network of Fig. below for the input shown.



5. What is significance of input and output resistance in a voltage amplifier.

6. Draw a negative clamper and explain its working along with the waveform

7. Draw the low frequency small signal model of a transistor in CE configurations and explain significance of each model.

8. What is Q point of operation of the transistor? What is the region of operation of the transistor when it is working as an amplifier?

9.Explain the significance of coupling and the bypass capacitors in a CE amplifier.

10. Draw the output characteristics of CE configuration and mark its regions of operation.

11.What is biasing? Explain its significance

12. Sketch the high frequency re model of a transistor and explain in brief.

13. Discuss the effect of coupling capacitors on low frequency response of BJT amplifiers.

14. Justify why the clamping circuit is also called a dc inserter?

15. Evaluate the equations for voltage gain , current gain, input impedance and output impedance for a BJT for CE configuration using re model.

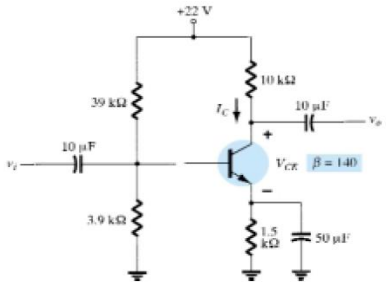
16. Evaluate the expression for Ri, Ai, Av and Ro for CE amplifier with un bypassed Re.

17. Explain the principle of clamping. What is the need for a shunting resistor R in parallel with the diode in the basic clamping circuit?

18. Estimate the gain of an amplifier at frequency = 100kHz ,if an Amplifier 3 dB gain is 200 and higher cut-off frequency is 20kHz.

19. For an amplifier, midband gain is 100 and lower cut-off frequency is 1kHz. Estimate the gain of an amplifier at frequency = 20Hz.

20. Determine the dc bias voltage V_{CE} and the current I_C for the voltage-divider configuration given below.



21. Determine DC bias voltage V_{CE} and current Ic for the self-bias circuit with V_{cc} =22V, R_c = 10KΩ, R_1 = 37KΩ, R_2 = 3.9KΩ, R_E = 1.5KΩ, β = 160, If the transistor is replaced by β = 50, Determine new Q points, Assume Silicon transistor. Comment on result .

22. A voltage divider bias CE amplifier has VCC = 14V , R1 =33KΩ, R2=6.8KΩ, RC= 1KΩ, RE =470Ω. The silicon transistor used has a β=100, CC1 =0.01µf,CC2= 0.1 µf , CE= 47µf. With the help of re model find the voltage gain, input impedance , output impedance.

23. Given the input signal shown in the figure (Vi=20sinwt) ,analyze and design a circuit to eliminate the noise , justify your answer.

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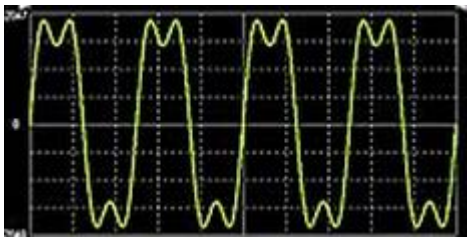
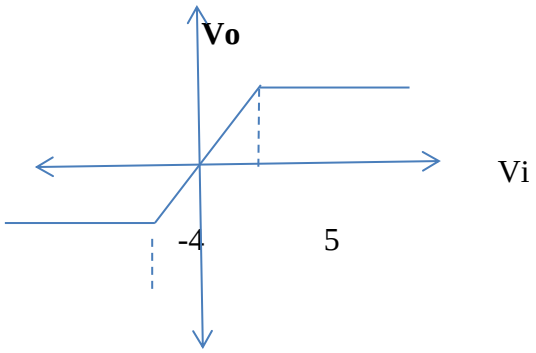
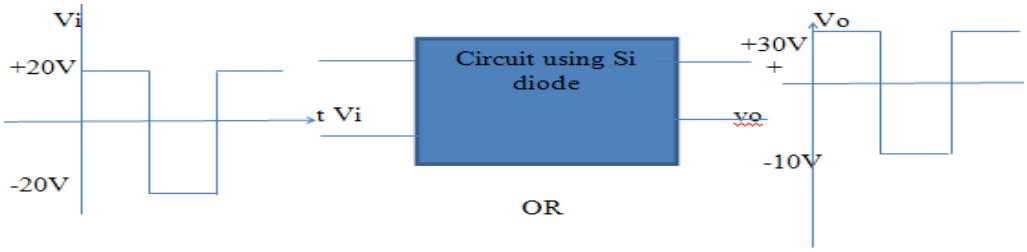


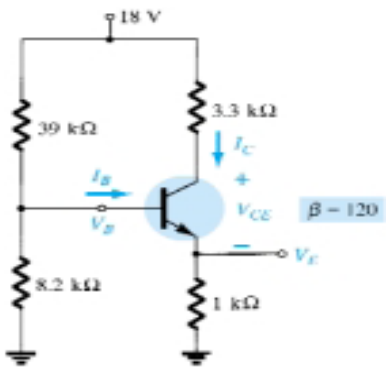
24. For the following transfer characteristics identify and analyze the circuit, also obtain the output wave form if the input wave form is 10sinwt.(assume Si diode)



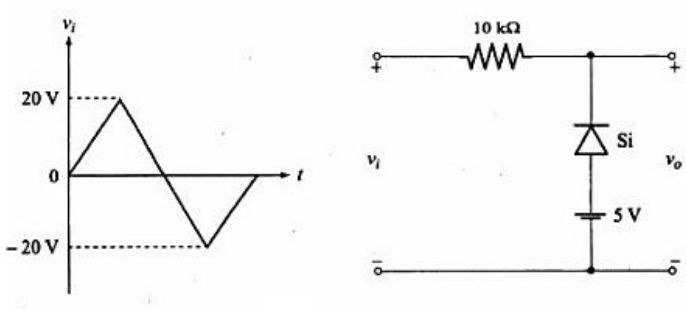
25. Design a suitable circuit for the block shown below which has input and output waveforms as indicated



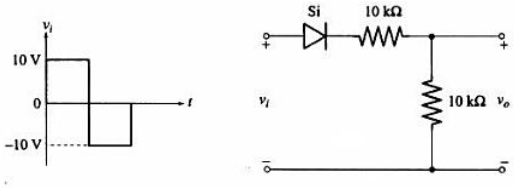
26. Determine the following for the voltage-divider configuration of Fig given below using the approximate approach: (a) I_C. (b) V_{CE}. (c) I_B. (d) V_E. (e) V_B.



27.For the circuit shown below, determine the transfer characteristics and sketch the output waveforms.

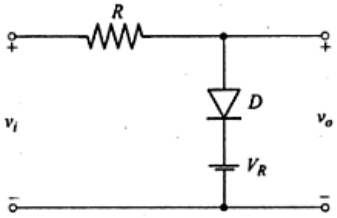


28.For the circuit shown below sketch the output waveform for the given input signal.

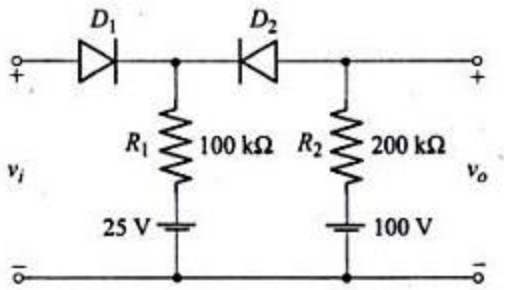


29.For the diode clipping circuit shown below, draw the input and output waveforms for:

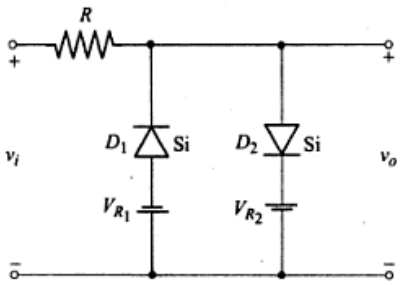
a) R=100Ω b) R=1KΩ c) R=10kΩ given v_i=20Sinωt and V_r=10V Assume r_f=100Ω,Rr=∞ and V_k=0



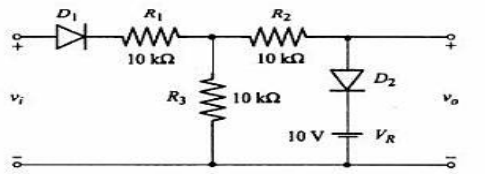
30.Sketch the output voltage waveform over the input voltage waveform for the circuit shown below given the input varies linearly from 0 to 150 V. Assume ideal diodes.



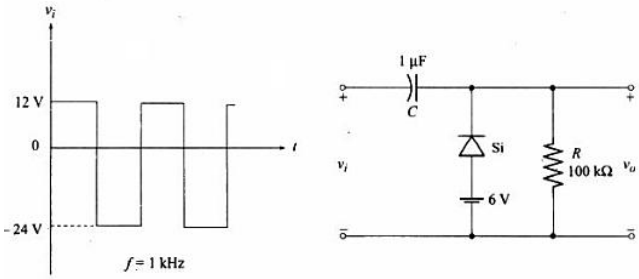
31. The circuit shown is used to square a 10kHz sinusoidal with a peak amplitude of 50V.Find Vr1 and Vr2, if it is desired that the output voltage be flat for 90% of the time. Assume rf=100Ω,Rr=100k Ω for the diodes. What is the reasonable value of R?



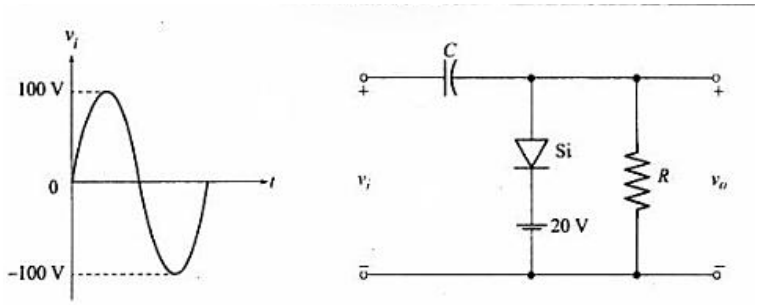
32.Plot the transfer characteristic of the circuit shown assuming ideal diodes and write the transfer characteristic equations. Sketch v0 if vi=20Sinωt.



33.For the circuit shown below, find and plot the waveform of vo for the input indicated.



34.For the circuit shown below, find and plot the waveform of vo for the input indicated assuming ideal diode.



35.The input to a clipping circuit is a sine wave of peak value 25V.Design the component values such that the output should have its positive peak clipped at15V and negative peak at 18V.

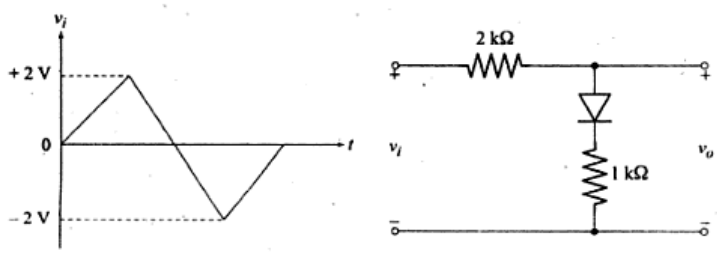
36.Design the values of R and C for a clamping circuit for which the input signal is a square wave of frequency 1KHz.

37. Design a clamping circuit to obtain an output with positive peak at 20 V and negative peak at -10V.The input is a square wave of ±15V at 1KHz.Assume silicon diode.

38.Sketch and explain the circuit of a double ended clipper using Si diodes which limits the output between ± 10V.

39. Sketch and explain the circuit of a double ended clipper using Si diodes which limits the output between 5V and 8V.

40.Assuming an ideal diode, draw the output voltage waveform for the input signal given below.



Questions On Feedback Amplifiers

1. What are the effects of negative feedback? Explain.

OR

1. Discuss the advantages and disadvantages of –ve feedback.

2.How do you classify the amplifiers based on input and output impedances ? Explain.

3.Write the block diagram of the following indicating A, Af and β .

a.Voltage series feedback amplifier

b.Voltage shunt feedback amplifier

c. Current series feedback amplifier

d.Current shunt feedback amplifier

4.Derive the expression for Af, Rif and Rof for the above amplifiers?

5. An amplifier without feedback has a transfer gain of -2000.The transfer gain changes by 20% due to temperature. If negative feedback with β=-0.1 is given, Calculate the change in the gain of the feedback amplifier? (Ans:1+A β =201,Change in gain=0.099%)

Solution: A=-2000, β=-0.1, 1+A β =201, dA_f/A_f= (dA/A)/(1+A β)=0.099%

6.Calculate the gain of a negative feedback amplifier having A=-2000, if the feedback factor is 20%?

(Ans: β=-0.2, 1+Aβ=401, Af= A/(1+Aβ)= -4.98)

7.Given A, β,Ri, Ro And BW, Find A_f, Rif, Rof and BWf for the following.

A.Voltage series feedback amplifier
B. Voltage shunt feedback amplifier
C. Current series feedback amplifier
D.Current shunt feedback amplifier

8. Draw the practical circuits for the following
A. Voltage series feedback amplifier
B. Voltage shunt feedback amplifier
C. Current series feedback amplifier

9.The following data is available for Voltage series feedback amplifier RB =50k ohm, RE =1k ohm, RC =1k ohm ,hie =2k ohm, hfe = 150 .Calculate

a. Av and Avf.

b. Ri and Rif.
c. Ro and Rof.

Write the circuit.

Ans: a) 75 & 0.986 b)1.923k ohm & 146.148 k ohm c)1 k ohm & 13.15 ohm **Solution**: (a)$A_V=h_{fe}R_E/h_{ie}$, $A_{Vf}=Av/(1+A\beta)$

(b) $R_i=R_B$ parallel hie, $Rif=Ri(1+A_V\beta)$

(c)$R_O=R_E=1K\Omega$,$R_O=R_O/(1+A\beta)$

10.For the current series feedback amplifier, calculate the following. Write the circuit.

a. Desensitivity Factor
b. Transfer gain with feedback.
c. Voltage Gain with & without feedback.
d. Input resistance with & without feedback.
e. Output resistance with & without feedback.

11. For transistor used hfe=200, hie=2k ohm, RB =390k ohm, RC =2k ohm, RE =560 ohm, Vcc=16V,Vs=10mV(rms).
(Ans: D=44.75, Af=-1.745m mho, Av=-156.25, Avf=-3.49, Ri=2.56k ohm, Rif=114.56k ohm, Ro=Rc=2k ohm, Rof=89.5k ohm).
Solution:
(a)$\beta= -R_E= - 560$ ohm
$A= - h_{fe}/(h_{ie}+R_E)$, $D=1+A\beta$

(b)$A_f=A/(1+A\beta)$

(c)$Av=A\ R_C$,$Av=A_f\ R_C$

(d)$R_i=h_{ie}+R_E$
$R_{if}= h_{ie}+ h_{fe}\ R_E$

(e)$R_O=R_C$
$R_{of}=R_O(1+A\beta)$
12. An amplifier without feedback has a voltage gain of 50, input resistance of 1 k Ω and output resistance of 2.5 kΩ. The input resistance of the current-shunt negative feedback amplifier using the above amplifier with a feedback factor of 0.2, is----------
13. A transistor current series feedback amplifier has the following data. RB=50KΩ, RC=2.7KΩ, RE=600Ω, VCC=15V, hfe=200, hie=1000 Ω. Determine AV and AVf. Write the practical circuit.

14. Calculate the harmonic distortion components for an output signal having fundamental amplitude of 2.5 V, second harmonic amplitude of 0.25 V, third harmonic amplitude of 0.1 V and fourth harmonic amplitude of 0.05 V.
Also calculate the total harmonic distortion.

Question Bank On Power Amplifiers.

1. How do you classify power amplifiers? Explain.

2.With a neat circuit diagram, waveforms and necessary equations explain the working of :

a) Series fed class A amplifier.
b) Transformer coupled class A power amplifier.
c)Transformer coupled class B push pull amplifier.
d) Complementary symmetry class B push pull circuit.

3. Derive an expression for conversation efficiency for
   a) Series fed class A power amplifier
   b) Transformer coupled class A power amplifier
   c) Class B power amplifier.

4. Why conversion efficiency is low in class A series fed circuit? How it is improved in transformer coupled circuit?

5.What is distortion in power amplifiers? Define harmonic distortion.

6.Define second harmonic and total harmonic distortion with necessary equations.

7.What is the need of heat sink? Explain.

8. Design a class B power amplifier to have an output power of 30 watt and an efficiency of 69%.Assume the missing data.

Question Bank on MOSFET

1. Differentiate between BJTs and FETs
2. Explain the physical structure of n – channel enhancement type MOSFET.

OR

Give the construction details of n-channel enhancement type MOSFET.

3. With a neat diagram and i_d- v_{ds} characteristics explain the operation of n – channel enhancement MOSFET. When
   (a)$V_{GS} \leq V_t$
   (b)$V_{GS} > V_t$ & $V_{DS} < V_{GS} - V_t$
   (c)$V_{GS} > V_t$ & $V_{DS} \geq V_{GS} - V_t$.
4. Define the following w.r.t n-channel enhancement MOSFET.

   (a)Threshold voltage.

   (b)Excess voltage or effective voltage or overdrive voltage.

   (c)Cutoff region

   (d)Tiode region

   (e)Saturation region.

   (f)Sub threshold region.

   (g)Process transconductance parameter

   (h)Aspect ratio

5. Derive an expression for i_D when the n-channel e-MOSFET operates in
   (a)Triode region.

(b)Saturation region.

6. Write a note on CMOS.
7. Explain the i_d -v_{GS} characteristics (transfer characteristics) of n-channel e-MOSFET in saturation region. Also write its large signal equivalent circuit model.
8. Write the simplified circuit symbol for
   (a)N-channel e-type MOSFET.
   (b)P- channel e-type MOSFET.
9. With a neat circuit diagram and i_d-V_{ds} characteristics ,explain the working of P-channel e-type MOSFET.
10. Explain the drain(i_d-v_{ds}) characteristics of CS amplifier.
11. Explain the drain(i_d-v_{ds}) and input-output characteristics of CS amplifier.
12. Define Q point & load line w.r.t MOSFET CS amplifier.
13. What is biasing? What are the types of biasing in MOSFET amplifiers?
14. With a neat circuit, explain a ---------- method of biasing technique. (any one method can be asked)

15.Explain how MOSFET is used as a switch.

16. EXAMPLE 4.1

Consider a process technology for which $L_{min} = 0.4\ \mu m$, $t_{ox} = 8$ nm, $\mu_n = 450\ cm^2/V \cdot s$, and $V_t = 0.7$ V.

(a) Find C_{ox} and k_n'.

(b) For a MOSFET with $W/L = 8\ \mu m/0.8\ \mu m$, calculate the values of V_{GS} and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100\ \mu A$.

(c) For the device in (b), find the value of V_{GS} required to cause the device to operate as a 1000-Ω resistor for very small v_{DS}.

Solution

(a)

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{8 \times 10^{-9}} = 4.32 \times 10^{-3}\ \text{F/m}^2$$

$$= 4.32\ \text{fF}/\mu\text{m}^2$$

$$k_n' = \mu_n C_{ox} = 450\ (\text{cm}^2/\text{V}\cdot\text{s}) \times 4.32\ (\text{fF}/\mu\text{m}^2)$$

$$= 450 \times 10^8\ (\mu\text{m}^2/\text{V}\cdot\text{s}) \times 4.32 \times 10^{-15} (\text{F}/\mu\text{m}^2)$$

$$= 194 \times 10^{-6}\ (\text{F/V}\cdot\text{s})$$

$$= 194\ \mu\text{A/V}^2$$

(b) For operation in the saturation region,

$$i_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_t)^2$$

Thus,

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} (V_{GS} - 0.7)^2$$

which results in

$$V_{GS} - 0.7 = 0.32\ \text{V}$$

or

$$V_{GS} = 1.02\ \text{V}$$

and

$$V_{DS\min} = V_{GS} - V_t = 0.32 \text{ V}$$

(c) For the MOSFET in the triode region with v_{DS} very small,

$$i_D \cong k_n' \frac{W}{L}(v_{GS} - V_t)v_{DS}$$

from which the drain-to-source resistance r_{DS} can be found as

$$r_{DS} \equiv \left.\frac{v_{DS}}{i_D}\right|_{\text{small } v_{DS}}$$

$$= 1\Big/\left[k_n' \frac{W}{L}(V_{GS} - V_t)\right]$$

Thus

$$1000 = \frac{1}{194 \times 10^{-6} \times 10(V_{GS} - 0.7)}$$

which yields

$$V_{GS} - 0.7 = 0.52 \text{ V}$$

Thus,

$$V_{GS} = 1.22 \text{ V}$$

EXAMPLE 4.2

Design the circuit of Fig. 4.20 so that the transistor operates at $I_D = 0.4$ mA and $V_D = +0.5$ V. The NMOS transistor has $V_t = 0.7$ V, $\mu_n C_{ox} = 100\ \mu A/V^2$, $L = 1\ \mu m$, and $W = 32\ \mu m$. Neglect the channel-length modulation effect (i.e., assume that $\lambda = 0$).

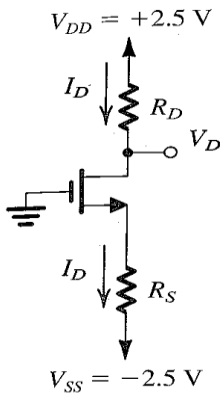


FIGURE 4.20 Circuit for Example 4.2.

Solution

Since $V_D = 0.5$ V is greater than V_G, this means the NMOS transistor is operating in the saturation region, and we use the saturation-region expression of i_D to determine the required value of V_{GS},

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_t)^2$$

Substituting $V_{GS} - V_t = V_{OV}$, $I_D = 0.4$ mA $= 400\ \mu$A, $\mu_n C_{ox} = 100\ \mu A/V^2$, and $W/L = 32/1$ gives

$$400 = \frac{1}{2} \times 100 \times \frac{32}{1} V_{OV}^2$$

which results in

$$V_{OV} = 0.5 \text{ V}$$

Thus,

$$V_{GS} = V_t + V_{OV} = 0.7 + 0.5 = 1.2 \text{ V}$$

Referring to Fig. 4.20, we note that the gate is at ground potential. Thus the source must be at -1.2 V, and the required value of R_S can be determined from

$$R_S = \frac{V_S - V_{SS}}{I_D}$$

$$= \frac{-1.2 - (-2.5)}{0.4} = 3.25 \text{ k}\Omega$$

To establish a dc voltage of +0.5 V at the drain, we must select R_D as follows:

$$R_D = \frac{V_{DD} - V_D}{I_D}$$

$$= \frac{2.5 - 0.5}{0.4} = 5 \text{ k}\Omega$$

EXAMPLE 4.3

Design the circuit in Fig. 4.21 to obtain a current I_D of 80 μA. Find the value required for R, and find the dc voltage V_D. Let the NMOS transistor have $V_t = 0.6$ V, $\mu_n C_{ox} = 200\ \mu A/V^2$, $L = 0.8\ \mu m$, and $W = 4\ \mu m$. Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).

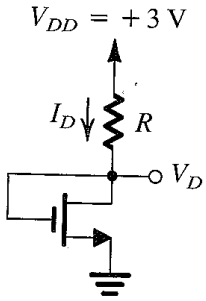


FIGURE 4.21 Circuit for Example 4.3.

Solution

Because $V_{DG} = 0$, $V_D = V_G$ and the FET is operating in the saturation region. Thus,

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_t)^2$$

$$= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{OV}^2$$

from which we obtain V_{OV} as

$$V_{OV} = \sqrt{\frac{2I_D}{\mu_n C_{ox}(W/L)}}$$

$$= \sqrt{\frac{2 \times 80}{200 \times (4/0.8)}} = 0.4 \text{ V}$$

Thus,

$$V_{GS} = V_t + V_{OV} = 0.6 + 0.4 = 1 \text{ V}$$

and the drain voltage will be

$$V_D = V_G = +1 \text{ V}$$

The required value for R can be found as follows:

$$R = \frac{V_{DD} - V_D}{I_D}$$

$$= \frac{3 - 1}{0.080} = 25 \text{ k}\Omega$$

EXAMPLE 4.4

Design the circuit in Fig. 4.22 to establish a drain voltage of 0.1 V. What is the effective resistance between drain and source at this operating point? Let $V_t = 1$ V and $k_n'(W/L) = 1$ mA/V^2.

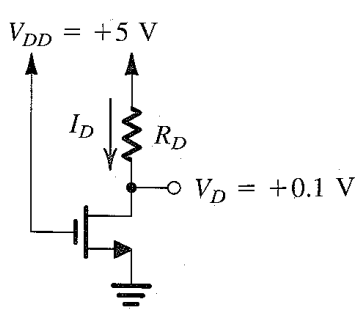


FIGURE 4.22 Circuit for Example 4.4.

Solution

Since the drain voltage is lower than the gate voltage by 4.9 V and $V_t = 1$ V, the MOSFET is operating in the triode region. Thus the current I_D is given by

$$I_D = k_n' \frac{W}{L}\left[(V_{GS} - V_t)V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

$$I_D = 1 \times \left[(5-1) \times 0.1 - \frac{1}{2} \times 0.01\right]$$

$$= 0.395 \text{ mA}$$

The required value for R_D can be found as follows:

$$R_D = \frac{V_{DD} - V_D}{I_D}$$

$$= \frac{5 - 0.1}{0.395} = 12.4 \text{ k}\Omega$$

In a practical discrete-circuit design problem one selects the closest standard value available for, say, 5% resistors—in this case, 12 kΩ; see Appendix G. Since the transistor is operating in the triode region with a small V_{DS}, the effective drain-to-source resistance can be determined as follows:

$$r_{DS} = \frac{V_{DS}}{I_D}$$

$$= \frac{0.1}{0.395} = 253 \ \Omega$$

EXAMPLE 4.5

Analyze the circuit shown in Fig. 4.23(a) to determine the voltages at all nodes and the currents through all branches. Let $V_t = 1$ V and $k_n'(W/L) = 1$ mA/V^2. Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).

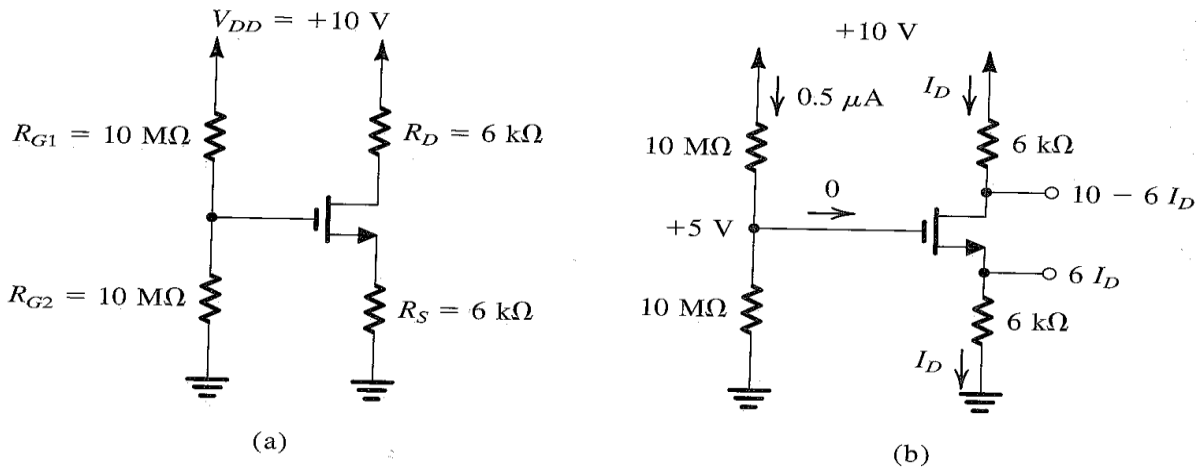


FIGURE 4.23 **(a)** Circuit for Example 4.5. **(b)** The circuit with some of the analysis details shown.

Solution

Since the gate current is zero, the voltage at the gate is simply determined by the voltage divider formed by the two 10-MΩ resistors,

$$V_G = V_{DD}\frac{R_{G2}}{R_{G2}+R_{G1}} = 10\times\frac{10}{10+10} = +5\text{ V}$$

With this positive voltage at the gate, the NMOS transistor will be turned on. We do not know, however, whether the transistor will be operating in the saturation region or in the triode region. We shall assume saturation-region operation, solve the problem, and then check the validity of our assumption. Obviously, if our assumption turns out not to be valid, we will have to solve the problem again for triode-region operation.

Refer to Fig. 4.23(b). Since the voltage at the gate is 5 V and the voltage at the source is I_D (mA) $\times$ 6 (kΩ) $= 6I_D$, we have

$$V_{GS} = 5 - 6I_D$$

Thus I_D is given by

$$I_D = \frac{1}{2}k_n' \frac{W}{L}(V_{GS} - V_t)^2$$
$$= \frac{1}{2} \times 1 \times (5 - 6I_D - 1)^2$$

which results in the following quadratic equation in I_D:

$$18I_D^2 - 25I_D + 8 = 0$$

This equation yields two values for I_D: 0.89 mA and 0.5 mA. The first value results in a source voltage of $6 \times 0.89 = 5.34$, which is greater than the gate voltage and does not make physical sense as it would imply that the NMOS transistor is cut off. Thus,

$$I_D = 0.5 \text{ mA}$$
$$V_S = 0.5 \times 6 = +3 \text{ V}$$
$$V_{GS} = 5 - 3 = 2 \text{ V}$$
$$V_D = 10 - 6 \times 0.5 = +7 \text{ V}$$

Since $V_D > V_G - V_t$, the transistor is operating in saturation, as initially assumed.

EXAMPLE 4.6

Design the circuit of Fig. 4.24 so that the transistor operates in saturation with $I_D = 0.5$ mA and $V_D = +3$ V. Let the enhancement-type PMOS transistor have $V_t = -1$ V and $k_p'(W/L) = 1$ mA/V^2. Assume $\lambda = 0$. What is the largest value that R_D can have while maintaining saturation-region operation?

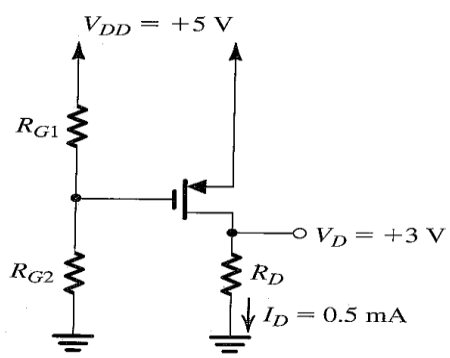


FIGURE 4.24 Circuit for Example 4.6.

Solution

Since the MOSFET is to be in saturation, we can write

$$I_D = \frac{1}{2} k_p' \frac{W}{L} (V_{GS} - V_t)^2$$

$$= \frac{1}{2} k_p' \frac{W}{L} V_{OV}^2$$

Substituting $I_D = 0.5$ mA and $k_p' W/L = 1$ mA/V^2 and recalling that for a PMOS transistor V_{OV} is negative, we obtain

$$V_{OV} = -1 \text{ V}$$

and

$$V_{GS} = V_t + V_{OV} = -1 - 1 = -2 \text{ V}$$

Since the source is at +5 V, the gate voltage must be set to +3 V. This can be achieved by the appropriate selection of the values of R_{G1} and R_{G2}. A possible selection is $R_{G1} = 2$ MΩ and $R_{G2} = 3$ MΩ.

The value of R_D can be found from

$$R_D = \frac{V_D}{I_D} = \frac{3}{0.5} = 6 \text{ k}\Omega$$

Saturation-mode operation will be maintained up to the point that V_D exceeds V_G by $|V_t|$; that is, until

$$V_{D_{\max}} = 3 + 1 = 4 \text{ V}$$

This value of drain voltage is obtained with R_D given by

$$R_D = \frac{4}{0.5} = 8 \text{ k}\Omega$$

EXAMPLE 4.9

It is required to design the circuit of Fig. 4.30(c) to establish a dc drain current $I_D = 0.5$ mA. The MOSFET is specified to have $V_t = 1$ V and $k_n'W/L = 1$ mA/V^2. For simplicity, neglect the channel-length modulation effect (i.e., assume $\lambda = 0$). Use a power-supply $V_{DD} = 15$ V. Calculate the percentage change in the value of I_D obtained when the MOSFET is replaced with another unit having the same $k_n'W/L$ but $V_t = 1.5$ V.

Solution

As a rule of thumb for designing this classical biasing circuit, we choose R_D and R_S to provide one-third of the power-supply voltage V_{DD} as a drop across each of R_D, the transistor (i.e., V_{DS}) and R_S. For $V_{DD} = 15$ V, this choice makes $V_D = +10$ V and $V_S = +5$ V. Now, since I_D is required to be 0.5 mA, we can find the values of R_D and R_S as follows:

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{15 - 10}{0.5} = 10\ \text{k}\Omega$$

$$R_S = \frac{V_S}{R_S} = \frac{5}{0.5} = 10\ \text{k}\Omega$$

The required value of V_{GS} can be determined by first calculating the overdrive voltage V_{OV} from

$$I_D = \tfrac{1}{2}k_n'(W/L)V_{OV}^2$$

$$0.5 = \tfrac{1}{2} \times 1 \times V_{OV}^2$$

which yields $V_{OV} = 1$ V, and thus,

$$V_{GS} = V_t + V_{OV} = 1 + 1 = 2\ \text{V}$$

Now, since $V_S = +5$ V, V_G must be

$$V_G = V_S + V_{GS} = 5 + 2 = 7\ \text{V}$$

To establish this voltage at the gate we may select $R_{G1} = 8$ MΩ and $R_{G2} = 7$ MΩ. The final circuit is shown in Fig. 4.31. Observe that the dc voltage at the drain (+10 V) allows for a positive signal swing of +5 V (i.e., up to V_{DD}) and a negative signal swing of –4 V [i.e., down to $(V_G - V_t)$].

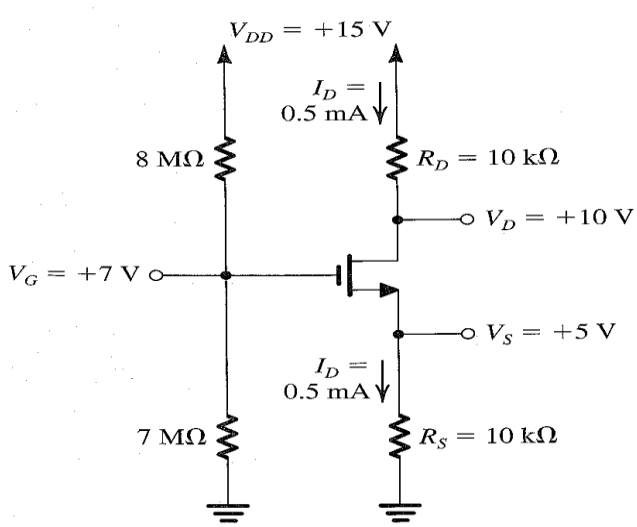


FIGURE 4.31 Circuit for Example 4.9.

If the NMOS transistor is replaced with another having $V_t = 1.5$ V, the new value of I_D can be found as follows:

$$I_D = \tfrac{1}{2} \times 1 \times (V_{GS} - 1.5)^2 \tag{4.47}$$

$$V_G = V_{GS} + I_D R_S$$

$$7 = V_{GS} + 10 I_D \tag{4.48}$$

Solving Eqs. (4.47) and (4.48) together yields

$$I_D = 0.455 \text{ mA}$$

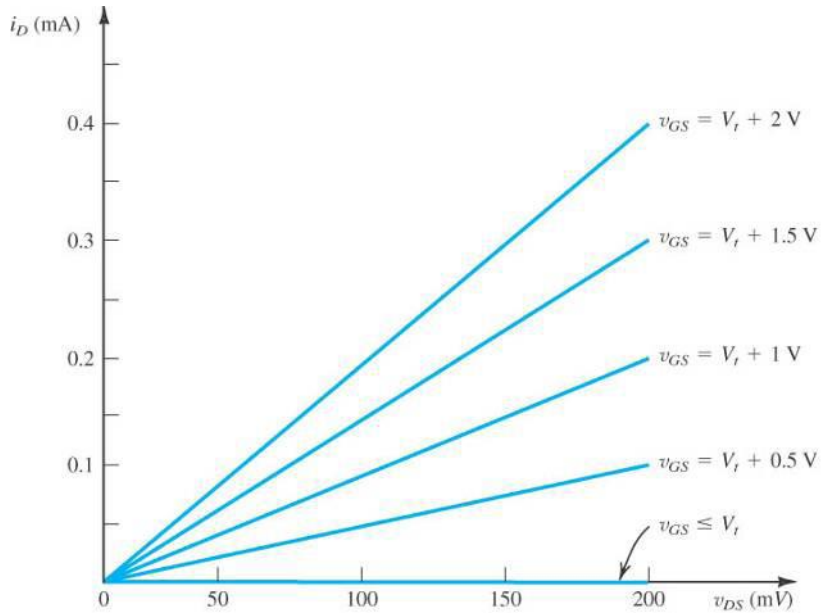
Thus the change in I_D is

$$\Delta I_D = 0.455 - 0.5 = -0.045 \text{ mA}$$

which is $\dfrac{-0.045}{0.5} \times 100 = -9\%$ change.

EXERCISE PROBLEMS ON UNIT-2 AND UNIT-3

4.1 From the description given below of the operation of MOSFET, for small V_{DS}, we note that I_D is proportional to $(V_{GS} - V_t)\, V_{DS}$. Find the constant of proportionality for the particular device whose characteristics are depicted as shown. Also give the drain to source resistances corresponding to the override voltage $(V_{GS}-V_t)$ of 0.5V to 2V.



Solution: a) $I_D \propto (V_{GS} - V_t)\ V_{DS}$

$$I_D = K\,(V_{GS} - V_t)\,V_{DS}$$

$$K = \frac{ID}{(VGS - Vt)\,VDS}$$

$$= \frac{0.4}{400}$$

1mA/V^2

b) Drain to source Impedance= 1/ slope of the graph

at $(V_{GS} - V_t) = 0.5V$

$$R_{DS}|_{0.5V} = \frac{1}{(\frac{ID}{VDS})}$$

$I_D = 200 \times 10^{-3}$, $V_{DS} = 0.1 \times 10^{-4}$

at(V_{GS} -V_t) =2V

$RDS|2V=\frac{VDS}{ID}$

I_D= 200 X 10^{-3} , V_{DS}=0.4 X 10^{-3}

=0.5kΩ

Range of drain to source Impedance= **2kΩ to 0.5KΩ** .

4.2 For a 0.8µm process technology, for which t_{ox}= 15nm and μ_n=550cm2/Vs, find Cox, kn' and the over drive voltage Vov=V_{GS}-V_t, required to operate a transistor having w/l =20 in saturation with I_D=0.2mA. What is the minimum value of V_{DS} needed?

Given: Lmin= 0.8 µm

t_{ox}= 15nm

µn= 550X $10^{-4}m^2V^{-1}s^{-1}$

w/l=20

I_D=200µA

To find C_{ox}:

$C_{ox}=\frac{\epsilon ox}{tox}$= 0.23 x$10^{-2}$Fm-2 (€ox= 3.45x 10-11 Fm-1)

To find kn':

$K_{n'}$= μ_n C_{ox}

= 126.5µAV^{-2}

To find V_{ov} in saturation region:

$I_D=\frac{Kn'}{2}\frac{w}{l}(V_{GS}-V_t)^2=\frac{Kn'}{2}\frac{w}{l}V_{ov}^2$

$V_{ov}=\sqrt{\frac{2IDL}{WKn'}}$= 0.397V

To find $V_{DS(min)}$:

$V_{DS(min)=}$ V_{GS}- V_t= V_{ov}=0.397V**= 0.40V**

4.3 Use the expression for operation in the triode region to show that n channel MOSFET operated with an override voltage $V_{ov}=V_{GS}-V_t$ and having a small V_{DS} across it behaves approximately as a linear resistance r_{DS}

$$R_{DS}=\frac{1}{kn'\frac{w}{L}Vov}$$

Calculate the value of r_{DS} obtained for a device having kn'=100μA/v^2 and w/L= 10 when operated with an overdrive voltage of 0.5V

Solution: WKT, resistance of drain to source is given by

$$R_{DS}=\frac{Vds}{ID}\Big|_{\text{small VDS}} \qquad (1)$$

In triode region, I_D can be expressed as

$$I_D=\frac{Kn'}{2}\frac{w}{l}\left[(V_{GS}-V_t)V_{DS}-\frac{Vds2}{2}\right]kn'\frac{w}{L}V_{ov}V_{DS} \qquad (2)$$

$$\frac{Vds}{ID}=\frac{1}{kn'\frac{w}{L}Vov}$$

Comparing (1) and (2)

$$R_{DS}=\frac{Vds}{ID}\Big|_{\text{small VDS}} = \frac{1}{kn'\frac{w}{L}Vov}$$

To calculate r_{DS}:

Given: kn'= 100μAV^{-2}

w/L = 10

V_{ov}=0.5

$$R_{DS}=\frac{1}{kn'\frac{w}{L}Vov}= 2k\Omega$$

4.4 An enhancement type NMOS transistor with V_t=0.7V has its source terminal grounded and a 1.5V dc applied to the gate. In what region does the device operate for

a) $V_{D=}$0.5V b) V_D= 0.9V c) $V_{D=}$3V

solution: The MOSFET operates in- triode region when $V_{DS}<V_{Ov}$ and in saturation region when $V_{DS}>V_{OV}$

here$V_{ov}=V_{GS}-V_t$= 1.5 0.7=0.8V

case a) when $V_D=V_{DS}$=0.5V is applied

$V_{DS}<V_{GS}-V_t$

Device operates in triode region.

Case b) when V_{DS}=0.9V

$V_{DS}> V_{GS} -V_t$

Device operates in saturation region.

Case c) when V_{DS}=3V

$V_{DS}> V_{GS}-V_t$

Device operates in saturation region.

4.5 If the NMOS device in 4.4 has $\mu_n C_{ox}$= 100µA/V^2 w= 10µm, L=1µm, find the value of the drain current that results in each of the three cases a) , b), c) specified in 4.4

Solution: a) device operates in triode region

$I_D= \frac{Kn'}{2}\frac{w}{l}$ [(V_{GS} -V_t)V_{DS}-$\frac{Vds2}{2}$]

Here, kn'=$\mu_n C_{ox}$= 100µAV^{-2}

W/L =10, $V_{GS}-V_t$=0.8V, V_{DS}=0.5V

$I_{D=}$275µA

b) Device operates in saturation region

$I_D = \frac{Kn'}{2}\frac{w}{l}(V_{GS} - V_t)^2$

$=0.5 \times 100\times10^{-6} \times 10(0.8)^2 = 320\mu A$

$I_D = 320\mu A$

c) device operates in saturation region

$I_D = \frac{Kn'}{2}\frac{w}{l}(V_{GS} - V_t)^2$

$I_D = 320\mu A$

4.6 An enhancement type NMOS transistor with V_t=0.7V conducts a current I_D=100µA when $V_{GS}=V_{DS}$=1.2V.Find the value of I_D for V_{GS}=1.5V and V_{DS}=3V. Also calculate the value of the drain to source resistance R_{DS} for small V_{DS} and V_{GS}=3.2V.

Solution: when $V_{GS}=V_{DS}$=1.2V and V_t=0.7V

$V_{DS} > V_{GS} - V_t$, device operates in saturation region.

Given: I_D=100µA

$I_D = \frac{Kn'}{2}\frac{w}{l}(V_{GS} - V_t)^2$

Substituting the values we get,

$= kn'\frac{w}{l} = 800 \times 10^{-6} AV^{-2}$

Now, V_{GS}=1.5V V_{DS}=3V

Again, $V_{DS} > V_{GS} - V_t$,saturation region

$I_D = \frac{Kn'}{2}\frac{w}{l}(V_{GS} - V_t)^2$

$=400 \times 10^{-6}(1.5-0.7)^2$

$=256 \times 10^{-6} A$

I_D= 256µA

To find R_{DS}:

$R_{DS}= \frac{1}{kn'\frac{w}{L}Vov}$ =500Ω

R_{DS}=500Ω

4.7 An NMOS transistor is fabricated in a 0.4µm process having $\mu_n C_{ox}$=200µA/V^2 and V'_A=50V/µm of channel length. If L=0.8µm and W=16µm, find V_A and Λ. Find the value of I_D that results when the device is operated with an overdrive voltage V_{OV}= 0.5V and V_{DS}=1V. Also, find the value of $r_o$at this operating point. If V_{DS} is increased by 2V, what is the corresponding change in I_D ?

Solution:

Given: $\mu_n C_{ox}$= kn'=200µAV^{-2}

V'_A= 50Vµm^{-2}

L=0.8µm

W=16µm

To find V_A: $V'_A=\frac{Va}{L}$

V_A= 50 x 0.8=40V

V_A=40V

To find Λ:

$\Lambda=\frac{1}{Va}$= 0.025V^{-1}

Λ=0.025V^{-1}

To find I_D: here $V_{DS}>V_{OV}$, saturation

$I_D= \frac{Kn'}{2}\frac{w}{l}(V_{GS} - V_t)^2$

I_D=0.5mA

To find r_o: $r_o=\frac{1}{\Lambda Id}$ =80kΩ

r_0= 80kΩ

4.8 The PMOS transistor shown in the figure has V_t=-1V ,K_p'=60μA/V^2 and W/L= 10.

a) Find the range of V_G for which the transistor conducts.

b) In terms of V_G find the range of V_D for which the transistor operates in the triode region.

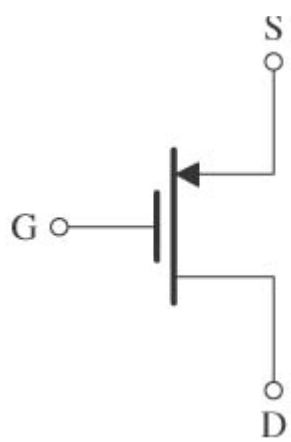
c) In terms of V_G find the range of V_D for which the transistor operates in the saturation region.

d) Neglecting channel length modulation (assuming Λ= 0) find the values of $|V_{OV}|$ and V_G and the corresponding range of V_D to operate the transistor in saturation mode with I_D= 75μA.

e) If Λ=-0.02V^{-1}, find the value of r_o corresponding to the overdrive voltage determined in d).

f) for Λ= -0.02V^{-1} and the value of V_{Ov} determined in d) find I_D and V_D=+3V and at V_D=0V; hence calculate the value of the apparent output resistance in saturation . compare to the value found in e).

V_S=5V



a) To induce a channel we apply gate voltage
$V_{GS} \leq V_t$
$V_G - V_S \leq V_t V_G \leq V_t + V_S$

Here V_t= -1V

V_s=5V

$V_G \leq 4V$

b) To operate in triode region

$V_{DS} \geq V_{GS}-V_t$

$V_D-V_S \geq (V_G-V_S)-V_t$

$V_D \geq V_G- V_t$

$V_D \geq V_G+1$ (V_t=-1V)

c) To operate in saturation region

$V_{DS} \leq V_{GS}-V_t$

$V_D-V_S \leq V_G-V_S-V_t$

$V_D \leq V_G- V_t$

$V_D \leq V_G+1$ (V_t=-1V)

d) In saturation mode,

$I_D= \frac{Kn'}{2}\frac{w}{l}V_{ov}^2(1+\Lambda V_{DS})$

$| V_{ov}^2|= 0.25$

$| V_{ov}| =0.5V$

$V_{ov}=V_{GS}-V_t$

$-0.5=V_{GS}+1$

$V_{GS}=-1.5V$

$V_{GS}=V_G-V_S$

$V_G=3.5V$

To operate in saturation,

$V_D \leq V_G+1$

$V_D \leq 4.5V$

e) $\Lambda=-0.02V^{-1}$

$V_A=\frac{1}{\Lambda} =50V$

$R_o = \frac{|Va|}{Id} = 0.667M\Omega$

f) At $V_D = +3V$

$V_D < 4.5$ MOSFET operates in saturation

$I_D = \frac{Kp'}{2}\frac{w}{l}V_{ov}^2(1+2\Lambda V_{DS})$

$V_{DS} = 3-5 = -2V$

$I_D = 78\mu A$

$R_o = \frac{|Va|}{Id} = 0.64M\Omega$

At $V_D = 0V$ ($V_D < 4.5V$) saturation

$I_D = \frac{Kp'}{2}\frac{w}{l}V_{ov}^2(1+\Lambda V_{DS})$

Substituting values

$I_D = 82.5\mu A$

4.15

$I_D = 0.32\,mA = \frac{1}{2} k_n' \frac{W}{L} V_{ov}^2 = \frac{1}{2} \times 1 \times V_{ov}^2 \Rightarrow V_{ov} = 0.8^V$

$V_{GS} = 0.8 + 1 = 1.8V$

$V_G = V_S + V_{GS} = 1.6 + 1.8 = 3.4^V$

$R_{G2} = \frac{V_G}{I} = \frac{3.4}{1\mu} = \underline{3.4\,M\Omega}$, $R_{G1} = \frac{5-3.4}{1\mu} = \underline{1.6\,M\Omega}$

$R_S = \frac{V_S}{0.32} = 5k\Omega$

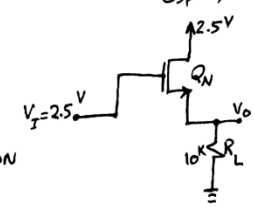
$V_{DS} \geq V_{ov} \Rightarrow V_D \geq V_{ov} + V_S \Rightarrow V_D \geq 0.8 + 1.6 = 2.4^V$

Assume $V_D = 3.4^V$, then $R_D = \frac{5-3.4}{0.32} = \underline{5k\Omega}$

4.16

$v_I = 0$: Since the circuit is perfectly symmetrical $\underline{V_O = 0}$ and therefore $V_{GS} = 0$ which implies the transistors are turned off and $\underline{I_{DN} = I_{DP} = 0}$.

$V_I = 2.5^V$: If we assume that the NMOS is turned on, then v_o would be less than 2.5^V and this implies that PMOS is off ($V_{GSP} > 0$)



$I_{DN} = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$

$I_{DN} = \frac{1}{2} \times 1 (2.5 - V_S - 1)^2$

$I_{DN} = 0.5 (1.5 - V_S)^2$

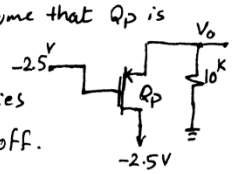
Also: $V_S = R_L I_{DN} = 10 I_{DN}$

$I_{DN} = 0.5 (1.5 - 10 I_{DN})^2$

$\Rightarrow 100 I_{DN}^2 - 32 I_{DN} + 2.25 = 0 \Rightarrow \underline{I_{DN} = 0.104\,mA}$

$I_{DP} = 0$, $V_o = 10 \times 0.104 = \underline{1.04V}$

$V_I = -2.5^V$: Again if we assume that Q_P is turned on, then $V_o > -2.5^V$ and $V_{GS1} < 0$ which implies the NMOS Q_N is turned off.



$I_{DN} = 0$

$I_{DP} = \frac{1}{2} k_n' \frac{W}{L} (V_{SG} - |V_t|)^2 = \frac{1}{2} \times 1 \times (V_S + 2.5 - 1)^2$

$V_S = -10 I_{DP} \Rightarrow 2 I_{DP} = (-10 I_{DP} + 1.5)^2$

$\Rightarrow \underline{I_{DP} = 0.104\,mA}$ $\Rightarrow V_o = -10 \times 0.104 = \underline{-1.04^V}$

4.17

a) $V_{IQ} = V_{GSQ} = 1.816V$ Refer to Example 4.8

$V_{IB} = 2V$

$V_{OQ} = 4^V$

$V_{OB} = 1^V$

b) By referring to Fig. 4.26c, negative peak of the output is $V_{OQ} - V_{OB} = 4 - 1 = \underline{3V}$ which corresponds to $V_{IB} - V_{IQ} = \underline{0.184^V}$ at the input.

c) The positive output peak is $V_{DD} - V_{OQ} = 10 - 4 = \underline{6^V}$ which corresponds to $V_{IQ} - V_t = 1.816 - 1 = \underline{0.816V}$ at the input.

d) From (b) and (c) the peak voltage at the inut is $\underline{0.184V}$ which translates to $\underline{3V}$ at the output. This is a gain of $\frac{3}{0.184} \simeq 16.3\,V/V$. This gain is different than the value obtained in example 4.8 ($14.7\,V/V$) due to the non-linear characteristic of the transfer function.

4.18

$v_o = V_{DD} - R_D i_D$

$i_D = \frac{1}{2} k_n' \frac{W}{L} (V_I - V_t)^2$ $\Big\} \Rightarrow v_o = V_{DD} - \frac{1}{2} R_D k_n' \frac{W}{L} (v_I - V_t)^2$ ①

Substitute $v_I = V_{IQ}$, $v_o = V_{OQ}$, $V_{ov} = V_{IQ} - V_t$

$\Rightarrow V_{OQ} = V_{DD} - \frac{1}{2} R_D k_n' \frac{W}{L} V_{ov}^2$

$\Rightarrow R_D k_n' \frac{W}{L} = \frac{2(V_{DD} - V_{OQ})}{(V_{ov})^2}$ ②

To derive an expression for A_v:

$A_v = \frac{\partial v_o}{\partial v_i}\Big|_{v_I = V_{IQ}} \overset{①}{=} -R_D k_n' \frac{W}{L} (\underbrace{V_{IQ} - V_T}_{V_{ov}})$

Substitute from ②:

$A_v = -\frac{2(V_{DD} - V_{OQ})}{V_{ov}^2} \times V_{ov} = -\frac{2(V_{DD} - V_{OQ})}{V_{ov}} = \frac{-2V_{RD}}{V_{ov}}$

If we calculate A_v for values in Exercise 4.17:

$A_v = -\frac{2(10-4)}{0.816} = -14.7\,V/V$

Same as the result in Example 4.8.

UNIT 5

1. Represent The Small Signal Model For An NMOS .
2. Derive The Expressions For Transconductance And Voltage Gain For Small Signal Operation Of The Enhancement MOSFET Amplifier.
3. Develop the Small-Signal Equivalent-Circuit Models To Show FET As A Voltage-Controlled Current Source.
4. Show The Three Different Relationships For Determining Gm .
5. Problem (4.30)
6. Develop The T Equivalent Circuit Model For The Mosfet.
7. Problem (4.69)
8. Analyse Small Signal Operation On A Common Source Amplifier, That Has A Signal Source At Its Input In Series With Rsig , And Load Rl Connected At Its Output Terminal. Also Develop an Equivalent Circuit Diagram To Determine The Overall Voltage Gain For The Configuratiojn.
9. Repeat The Q(8) For Common – Source Amplifier With A Source Resistance.
10. Explain Ic Biasing Circuits And Its Operation.
11. Derive The Output Current Equation For A Constant Current Source Mosfet.
12. Explain The Output Characteristics Of A Current Mirror Circuit.
13. Explain With Equations A Mos Current-Steering Circuit.
14. Analyse a Basic Bjt Based current Mirror Circuit.
15. Determine The Current Transfer Ratio Of A Wilson Current Mirror.
16. Prob (6.24)

$W_2 = 4W_1 \quad L_1 = L_2 \quad V_{ov} = 0.3V \quad , I_{REF} = 20\mu A$

$I_o = I_{REF} \dfrac{(W/L)_2}{(W/L)_1} = 20 \times 4 = \underline{80\mu A}$

$v_o\,min = V_{ov} = \underline{0.3V}$

$v_t = 0.5V$. According to Eq. 6.11 $I = \dfrac{(W/L)_2}{(W/L)_1} I_{REF} \left(1 + \dfrac{V_o - V_{GS}}{V_{A2}}\right)$

$v_{ov} = V_{GS} - V_t \Rightarrow V_{GS} = 0.3 + 0.5 = 0.8V$

$1 + \dfrac{V_o - V_{GS}}{25} = 1 \Rightarrow \underline{V_o = 0.8V}$

Or we could simply say $V_{DS1} = V_{DS2} = V_o$ and

Since $v_{DS1} = v_{GS1} = 0.8^V \Rightarrow \underline{v_o = 0.8^V}$

$$r_{o2} = \frac{V_A}{I_{D2}} = \frac{25}{0.08} = 312.5\,k\Omega$$

$$r_{o2} = \frac{\Delta v_o}{\Delta I_o} = \frac{1}{\Delta I_o} \Rightarrow \Delta I_o = \frac{1}{312.5^k} = \underline{3.2\,\mu A}$$

17. Prob (4.82)

Refer to Eq. 4.90: $G_v = \frac{R_G}{R_G + R_{sig}} \frac{g_m (R_D \| R_L)}{1 + g_m R_s}$

$G_{v1} = -16\ V/V$ reduced by factor of 4:

$$\frac{G_{v2}}{G_{v1}} = \frac{1}{1 + g_m R_s} \Rightarrow \frac{1}{4} = \frac{1}{1 + 2R_s} \Rightarrow \underline{R_s = 1.5\,k\Omega}$$

18. Characterize And Compare Non unilateral Amplifiers With Transistor Amplifiers.
19. Prob (4.30), Exercise
20. Example (4.10)